

Matthew Topham

AI Systems & Full-Stack Developer

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Computer science student at CU Boulder building full-stack AI systems for interactive storytelling, wearable context, and local inference infrastructure. I work across product interfaces, model integration, persistent state, local compute, and automated validation.

OPEN-SOURCE EXPERIENCE

Marinara Engine + Agents · Pasta Devs

Open-Source Developer · 2026-Present

- Contributed 100 merged pull requests across Marinara Engine (68) and Agents (32), spanning React/TypeScript UI, Node.js services, persistent world systems, agent capabilities, multimodal generation, and automated tests.
- Implemented a working MiniMax H3 / ComfyUI roleplay-animation pipeline: narration becomes a storyboard, character sheets/personas and World Maps artwork ground each scene, generated keyframes anchor the shots, and the final motion plan drives image-to-video generation.
- Live-tested the H3 flow on a successful multi-shot clip with synchronized motion and audio; packaged the proven prompt chain as reusable Storyboard choices.
- Built World Maps/spatial state, reusable character and location references, versioned host APIs for downloadable agents, and recovery paths for stateful workflows.

Selected work: Animation pipeline · MiniMax H3 prompts · World Maps · Agent capability APIs

SELECTED PROJECTS

Manfred · Wearable AI memory and context

Creator · 2026 · Prototype

- Built an Android/Flutter companion and Python services for Omi wearable audio ingestion, daily transcription/lifelogging, searchable notes, and durable upload buffering.
- Connected E09 smart glasses to ChatGPT Live for voice interaction and image uploads through phone automation; voice works today while photo delivery is still being hardened.

Demerzel · Personal AI operations and local fleet

Creator · 2026

- Built tooling for daily AI briefs, LLM wiki ingestion from daily conversations, and management of local AI services across an AMD Radeon AI PRO R9700, dual RTX 3060, and single RTX 3060 systems.
- Developed service health checks and workload controls to start/stop supported jobs and avoid GPU contention; separate autonomous coding/research workers remain under refinement.

School of the Ancients · AI / VR learning

Creator / Developer

Built conversational learning experiences with AI historical mentors across web and VR prototypes using React, TypeScript, Node.js, Supabase/Postgres, real-time voice AI, and Unity.

TECHNICAL SKILLS

Software: Python, TypeScript, JavaScript, SQL, React, Node.js, Flutter, REST, WebSockets. AI / infrastructure: local inference, ComfyUI, Linux, Docker, GitHub Actions, Playwright, GPU workload management, regression testing.

EDUCATION & CERTIFICATION

University of Colorado Boulder — B.A. Computer Science, expected May 2027

Front Range Community College — A.S. Computer Science, 2025

NVIDIA-Certified Associate — AI Infrastructure & Operations, August 2025